

Step-RL v2.0

>
**v2.0 | **20265 | **



Step-RL v2.0 Web Agent LLM, Large Language Model Agent
Grounding Hallucination
Evidential
Learning Curriculum
Scheduler MinHash PPO Proximal Policy
Optimization/GRPO Group Relative Policy Optimization KL
Step-RL v2.0 58% 86~91% 57% 87.5%
95.8% 24.5 11.5~13.2 46% 32% 4~6% MinHash
GRPO Value Model 4-bit 6~7 GB VRAM PPO 30%
RTX 4060 7B

1.

1.1

1.1.1 Web Agent

LLM, Large Language Model LLM Agent Agent Web
Long-Horizon Task 10~30 LLM Agent
Sparse Terminal Reward Credit
Assignment
Action Grounding
Hallucination LLM
Early Error Accumulation
SFT, Supervised Fine-Tuning Prompt RL, Reinforcement
Learning LLM
Agent Web RL

1.1.2 LLM Agent

PPO, Proximal Policy Optimization GRPO, Group Relative Policy
Optimization RL LLM Alignment LoRA, Low-Rank

Adaptation**PEFT, Parameter-Efficient Fine-Tuning**WebAge
ntRLTRLWeb

1.2

Step-RL v2.0**WebAgent**58%**86%~9
1%**≥95.8%**11.5~13.2**

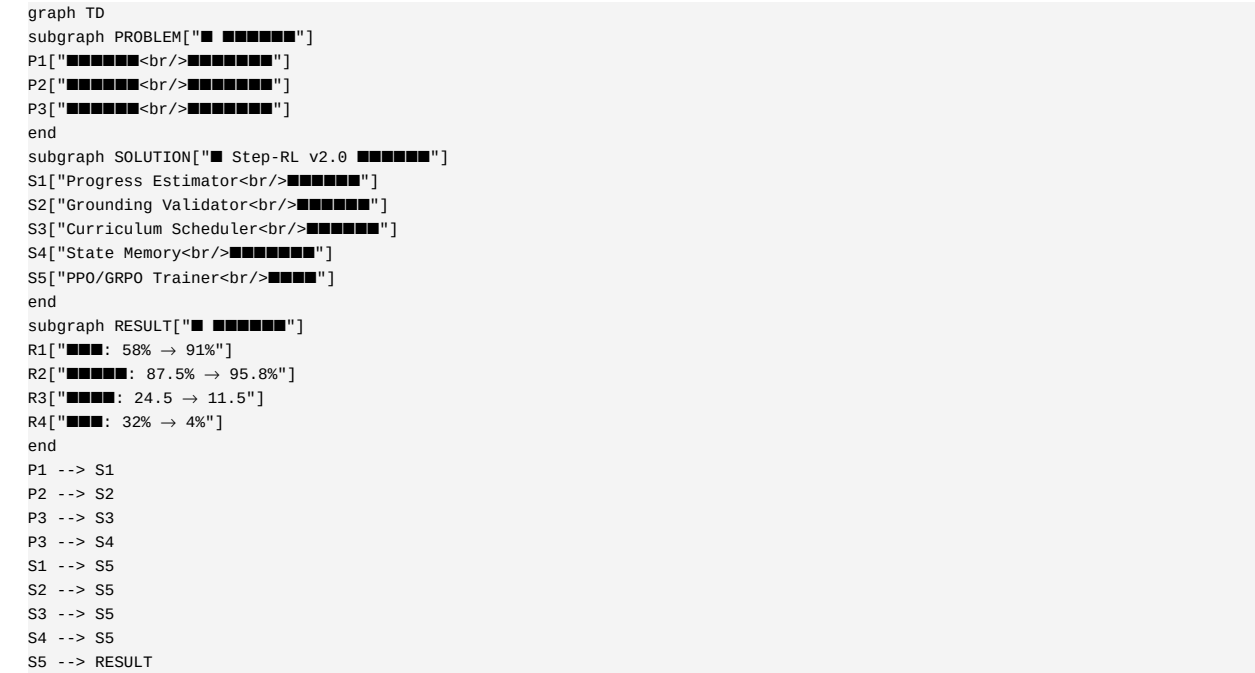
- **Progress Estimator
- **Grounding Validator
- **Curriculum Learning
- **
- **PPO/GRPO**GRPOValue Model30%8GB VRAM

1.3

**Step-RL v2.0

- **
- **
- **
- **DockerCI/CD
- **
- **

v1.0v2.0



1-1Step-RL v2.0--

```
graph LR
subgraph INPUT["■■■■"]
TASK["■■■■<br/>■■■■"]
end
subgraph CURRICULUM["■■■■"]
CS["Curriculum Scheduler<br/>■■■■ + ■■■■"]
end
```

```

subgraph AGENT["Agent"]
POL["Qwen3-8B + LoRA"]
end
subgraph VALIDATION[""]
GV["Grounding Validator"]
end
subgraph ENV["Web"]
PW["Playwright Chromium"]
end
subgraph REWARD[""]
PE["Progress Estimator"]
SM["State Memory"]
SUM[" $R_{total} = \alpha \cdot r_p + \beta \cdot r_g + \dots$ "]
end
subgraph TRAIN[""]
RL["PPO / GRPO + LoRA"]
end
TASK --> CS
CS --> POL
POL -->|Action JSON| GV
GV -->|/| PW
PW -->|Observation + State| PE
PE -->|State Hash| SM
SM -->|r_progress| SUM
GV -->|r_grounding| SUM
SUM -->|r_loop + r_novelty| SUM
SUM -->|Total Reward| RL
RL -->|/| POL

```

2-1 Step-RL v2.0

2.2

2.2.1

Step-RL v2.0 Web Action Space

- `click`
- `type`
- `scroll`/
- `goto`URL
- `wait`
- `finish`

/ / App

2.2.2

NFR, Non-Functional Requirement

2-2 Step-RL v2.0

GRPO 4-bit 6~7GB RTX 4060 8GB VRAM 7B

2.3

2.3.1

Step-RL v2.0

- `**Hugging Face**Qwen3-8B-Instruct / Qwen2.5-7B-Instruct` fallback
- `**Playwright**WebChromiumJS DOM`mcr.microsoft.com/playwright/python:v1.43.0-jammy``
- `**iPhone 15**Episode`

2.3.2

`**Gradio Demo7860**WebAgent`

`**FastAPI8000**RESTful API`

`**SFT LoRA Adapter`./outputs/sft_adapter/`Progress Estimator`./checkpoints/progress_estimator/`RLCheckpoint`./checkpoints/`./outputs/benchmark/``

```
graph TB
    subgraph UPSTREAM[""]
        U1["Hugging Face<br/>Qwen3-8B-Instruct"]
        U2["Playwright<br/>Chromium v1.43+"]
        U3["<br/>"]
    end
    subgraph STEPRV["Step-RL v2.0"]
        CORE["<br/> . . . "]
    end
    subgraph DOWNSTREAM[""]
        D1["Gradio Demo<br/> 7860"]
        D2["FastAPI <br/> 8000"]
        D3["Checkpoint / Adapter<br/>./checkpoints/ ./outputs/"]
    end
    U1 -->|CORE| CORE
    U2 -->|CORE| CORE
    U3 -->|CORE| CORE
    CORE -->|D1| D1
    CORE -->|API| D2
    CORE -->|D3| D3
```

2-2Step-RL v2.0

3.

Step-RL v2.0 YML

3.1

3.1.1

Single Responsibility Principle, Python

SRPEnvironmentRewardPolicyTrainingEvaluation

[illegible]

3.1.2 ■■■■

```

#####**YAML          #####YAML          Configuration-Driven#####          #####`config.yaml`
#####
grounding##### `CurriculumScheduler.get_reward_weights`() #####
##### YAML #####

```

3.1.3 ■■■■

```

##### validate_url() ##### Sandbox Domain Filtering ##### `urlparse` #####
hostname##### `localhost`#####`127.0.0.1`#####`file://`##### `escape_css_string()` #####
`escape_xpath_string()` ##### Escaping##### CSS/XPath ##### Dockerfile ##### root
#####`USER appuser`#####`torch.load(..., weights_only=True)`##### pickle #####

```

3.2 ■■■

[illegible]

3.2.1 ■■■

```
██████████Web ██████████`PlaywrightWebEnv`  
██████████`GroundingValidator` ██████████`Locato  
r` █`locator.py` ██████████Multi-Attribute Cascade Matching ██████████ `element_id >  
element_text(+tag) > xpath > css_selector > coordinates` ██████████
```

3.2.2 ■■■

```

#####`Agent`#####**Qwen3-8B-Instruct**#####LLM#####**LoRA**Low-Rank Adaptation#####**
Adapter##### 0.1% #####`ProgressEstimator`##### LLM Encoder ##### 3 ##### 512 ##### MLP
##### `[0,1]`#####**Evidential Learning**##### Dirichlet #####`StateMemory`#####
**MinHash**#####**LRU**Least Recently Used#####**##### 500 #####

```

3.2.3 ■■■

[illegible][illegible]

3.2.4 ■■■

BaseTrainer` Abstract Base Class, ABC rollout prompt
PPO GRPO 80% PPOTrainer` BaseTrainer` **GAE Generalized Advantage
Estimation** Policy + Reference + Value KL GRPOTrainer`
BaseTrainer` **Group-Relative Policy Optimization** Value Model 30%
last-token log-prob rollout update

3.2.5

JSON/JSONL JSON
ID Hugging Face Safetensors PyTorch .pt LoRA
Adapter CSV PNG

```
flowchart TB
    subgraph [" (Environment Layer)"]
        A1[PlaywrightWebEnv<br/>]
        A2[GroundingValidator<br/>]
        A3[Locator<br/>]
    end

    subgraph [" (Business Layer)"]
        B1[Agent<br/>Qwen3-8B + LoRA]
        B2[ProgressEstimator<br/>]
        B3[StateMemory<br/>]
    end

    subgraph [" (Control Layer)"]
        C1[CurriculumScheduler<br/>]
        C2[ContinualLearner<br/>]
        C3[Reward Summation<br/>]
    end

    subgraph [" (Training Layer)"]
        D1[BaseTrainer<br/>]
        D2[PPOTrainer<br/>GAE + KL]
        D3[GRPOTrainer<br/>]
    end

    subgraph [" (Data Layer)"]
        E1[JSON/JSONL]
        E2[JSON]
        E3[Safetensors/PT]
        E4[CSV/PNG]
    end

    A1 --> B1
    A2 --> B1
    A3 --> A2
    B1 --> C3
    B2 --> C3
    B3 --> C3
    C1 --> C3
    C2 -. -> B2
    C3 --> D1
    D1 --> D2
    D1 --> D3
    D2 --> E1
    D3 --> E1
    D2 --> E3
    D3 --> E3
    E1 --> E4
    E2 -. -> E2
```

3.3

BaseTrainer.collect_rollouts() rollout CurriculumScheduler.sample_task()
PlaywrightWebEnv.reset() Agent JSON
GroundingValidator.validate_and_correct() Jaccard bigram
wait PlaywrightWebEnv.execute_action() StateMemory.compute_hash() MinHash

update() ProgressEstimator (1 - uncertainty)
Reward Summation Trajectory Rollout 25%

```
flowchart LR
    subgraph Rollout["Rollout "]
        R1[Curriculum<br/>] --> R2[Env<br/>reset]
        R2 --> R3[Agent<br/>]
        R3 --> R4[Grounding<br/>]
        R4 --> R5[Env<br/>]
        R5 --> R6[StateMemory<br/>]
        R6 --> R7[ProgressEstimator<br/>+ ]
        R7 --> R8[Reward Summation<br/>]
        R8 --> R9[done?]
        R9 --> R10[Trajectory<br/>]
    end

    subgraph Update["Update "]
        U1[Replay Buffer<br/>25%] --> U2[BaseTrainer<br/>PPO/GRPO]
        U2 --> U3[Checkpoint<br/>]
    end

    R10 --> U1
```

3.4

3.4.1

** Qwen3-8B-Instruct Tool Use 8B 8GB
4-bit Qwen2.5-7B/14B

** Playwright Selenium Playwright API async/await Python Chromium
Headless Mode Selenium WebDriver page.evaluate() JavaScript DOM
Selenium accessibility API Playwright
route.abort() CSS

3.4.2 RL

**RL PPO GRPO PPO Policy + Reference + Value GAE FP16
24GB GRPO Value Model Policy + Reference FP16
16GB 4-bit 6-7GB 8GB VRAM GRPO $A_i = (R_i - \text{mean}(R_{\text{group}})) / (\text{std}(R_{\text{group}}) + \epsilon)$

** PEFT/LoRA Parameter-Efficient Fine-Tuning $r=64, \text{lora_alpha}=32$
q_proj/k_proj/v_proj/o_proj/gate_proj/up_proj/down_proj 0.1% 8B 80 LoRA
60% Adapter MB

** GRPO 8GB GPU LoRA

3.5

rollout

3.6.1 v1.0

v1.0 PPO Value Model 8GB
OOMOut of MemoryAgent Experience Replay
rollout

3.6.2 v2.0

v2.0 `PPOTrainer` `BaseTrainer`ABC`PPO
GRPO `update()` 80%`locator.py` `PlaywrightWebEnv`
`GroundingValidator` `CurriculumScheduler`
`StateMemory` MinHash LRU `ContinualLearner` GRPO
8GB GPU

`**v2.0`

Step-RL v2.0

4

4.1

Step-RL v2.0 Layered DecouplingWeb
AgentData Contract

4.2 PlaywrightWebEnv

4.2.1

``PlaywrightWebEnv`` [Playwright](https://playwright.dev/)
``async`/`await``
`Observation` HTML
`Step-RL v2.0`
JavaScript DOM `tag` `role` `text` `id` `coords` Playwright
≥ 1.60 `page.accessibility` API JS BeautifulSoup

```
# step_rl/environment/playwright_env.py lines 234-277
js_code = """
() => {
  const results = [];
  const tags = ['a', 'button', 'input', 'textarea', 'select', 'label',
'h1', 'h2', 'h3', 'h4', 'h5', 'h6', 'p', 'span', 'div',
'li', 'td', 'th'];
```

```

tags.forEach(tag => {
  document.querySelectorAll(tag).forEach((el, idx) => {
    if (el.offsetParent === null && tag !== 'div') return;
    const rect = el.getBoundingClientRect();
    const text = (el.innerText || el.textContent || el.value || el.placeholder || '').trim();
    const role = el.getAttribute('role') || tag;
    const id = el.id || el.getAttribute('data-testid') || '';
    if (text.length > 0 || id.length > 0 || tag === 'input' || tag === 'button' || tag === 'a') {
      results.push({
        tag: tag, role: role, text: text.slice(0, 200), id: id,
        coords: `(${Math.round(rect.x)}, ${Math.round(rect.y)})`,
        visible: el.offsetParent !== null
      });
    }
  });
});
return results;
}
"""
try:
  elements = await page.evaluate(js_code)
except Exception as e:
  logger.warning(f"JS extraction failed: {e}, falling back to BeautifulSoup")
html = await page.content()
from bs4 import BeautifulSoup
soup = BeautifulSoup(html, "lxml")
text = soup.get_text(separator="\n", strip=True)

```

```

**Security Sandbox validate_url blocked_domains
localhost 127.0.0.1 Action Space
click type scroll goto wait finish

```

4.2.2

```

execute_action Action click type robust_locate
Post-Action SPA Wait wait_for_load_state
domcontentloaded networkidle SPA

```

4.3 GroundingValidator

4.3.1

```

GroundingValidator DOM → visible + enabled
+ bounding_box → click type
Multi-Attribute Cascade Matching

```

```

element_id > element_text + tag > xpath > css_selector > coordinate_fallback ID
coordinates locator.py

```

```

PlaywrightWebEnv GroundingValidator

```

```

# step_rl/environment/grounding_validator.py lines 70-125
async def validate(self, page, action, params):
  if action in ("wait", "finish"):
    return GroundingResult(valid=True, reward=0.0)
  if action == "goto":
    url = params.get("url", "")
    if not url.startswith(("http://", "https://", "about:")):
      return GroundingResult(valid=False, reward=self.reward_failed, corrected_action=self._wait_action())
    return GroundingResult(valid=True, reward=self.reward_valid)
  if action == "scroll":
    return GroundingResult(valid=True, reward=self.reward_valid)
  # For click / type: need element
  locator, match_info = await robust_locate(page, params, multi_attribute_match=self.multi_attribute_match)
  if locator is not None:

```


4.5 StateMemory

4.5.1 MinHash

``StateMemory`` State Aliasing URL DOM
``StateSignature`` URL DOM MinHash Deterministic
MinHash precomputed permutations word
shingles

```
# step_rl/memory/state_memory.py lines 81-123
def _minhash(self, text, url, num_perm=64):
    words = text.lower().split()
    if len(words) < 2:
        return self._simple_hash(text, url)
    shingles = set()
    for i in range(len(words) - 1):
        shingles.add(f"{words[i]} {words[i+1]}")
    p = (1 << 61) - 1
    hashes = []
    for seed in range(num_perm):
        a = (seed * 1234567891 + 1) & 0xFFFFFFFFFFFFFFFF
        b = (seed * 9876543211 + 1) & 0xFFFFFFFFFFFFFFFF
        min_val = p
        for s in shingles:
            h = int(hashlib.md5(s.encode()).hexdigest(), 16)
            perm = ((h * a + b) & 0xFFFFFFFFFFFFFFFF) % p
            if perm < min_val:
                min_val = perm
        hashes.append(min_val)
    band_size = 4
    bands = []
    for i in range(0, num_perm, band_size):
        band = tuple(hashes[i : i + band_size])
        band_hash = hashlib.md5(str(band).encode()).hexdigest()[:4]
        bands.append(band_hash)
    url_h = hashlib.md5(url.encode()).hexdigest()[:8]
    return f"mh-{url_h}-{ '-' .join(bands)}"
```

hashlib.md5 shingle Hierarchical
Hashing

4.5.2

``StateMemory`` loop_window = 3 loop_penalty_base = -0.1
Novelty

Bonus

LRU Least Recently Used OrderedDict popitem(last=False)
max_states = 500

4.6 CurriculumScheduler

4.6.1

``CurriculumScheduler`` Curriculum Learning L1 L2 L3 L4
50% L1 + 40% L2 + 10% L3
10% L1 + 10% L2 + 40% L3 + 40% L4

Promotion Logic 20 `≥` 90% 10
epoch

CurriculumScheduler	$\beta = 2.0$	Grounding
Reward	$\alpha = 2.0$	Progress Reward
	$\epsilon = 0.2$	Phased Reward
Shaping		

XX

Component	Step-RL (Black)	v2.0 (Grey)
PlaywrightWebEnv	~100	~1000
GroundingValidator	~100	~1000
ProgressEstimator	~100	~1000
StateMemory	~100	~1000
MinHash	~100	~1000
LRU	~100	~1000
CurriculumScheduler	~100	~1000

— — —

```

class ProgressEstimator {
+ProgressOutput
+EvidentialLayer
+encode_observation(ids, mask, step) Tensor
+forward(ids, mask, step) ProgressOutput
+mc_dropout_predict(ids, mask, step) ProgressOutput
}
class StateMemory {
+StateSignature
+compute_hash(text, url, screenshot) str
+update(state_hash) (float, float, Dict)
+is_looping(state_hash, threshold) bool
}
class CurriculumScheduler {
+Task
+sample_task(epoch) Task
+get_reward_weights(epoch) Dict
+record_episode_result(level, success) None
}
PlaywrightWebEnv --> Locator : ■■
GroundingValidator --> Locator : ■■
PlaywrightWebEnv --> GroundingValidator : ■■■■
ProgressEstimator ..> StateMemory : ■■■■■■
CurriculumScheduler ..> PlaywrightWebEnv : ■■■■■■

```

```

sequenceDiagram
autonumber
participant Agent as Agent / Trainer
participant Env as PlaywrightWebEnv
participant GV as GroundingValidator
participant Loc as Locator
participant Page as Browser Page

Agent->>Env: execute_action(Action)
Env->>GV: validate(page, action, params)
GV->>Loc: robust_locate(page, params, multi_attribute=True)
Loc->>Page: element_id / text / xpath / css / coords
Page-->>Loc: Locator | None
Loc-->>GV: (locator, match_info)
alt Element Found & Interactive
GV->>Page: _check_interactivity()
Page-->>GV: ok=True
GV-->>Env: valid=True, reward=+0.1
Env->>Page: click / type / scroll
Page-->>Env: DOM updated
else Not Interactive
GV->>Loc: _find_similar_interactive()
alt Jaccard ≥ 0.85
Loc-->>GV: candidate.similarity ≥ 0.85
GV-->>Env: corrected_action, reward=-0.05
else No Match
GV-->>Env: wait fallback, reward=-0.2
end
end
Env->>Page: _wait_for_page_ready()
Page-->>Env: domcontentloaded / networkidle
Env-->>Agent: StepResult(obs, reward, done, info)

```

```

stateDiagram-v2
[*] --> L1_SinglePage : ■■■
L1_SinglePage : ■■■■ (2-3■)
L2_CrossPage : ■■■■ (4-7■)
L3_ComplexForm : ■■■■ (8-15■)
L4_MultiGoal : ■■■■ (15-30■)

L1_SinglePage --> L2_CrossPage : ■■■≥90% & ■■≥10
L2_CrossPage --> L3_ComplexForm : ■■■≥90% & ■■≥10
L3_ComplexForm --> L4_MultiGoal : ■■■≥90% & ■■≥10
L4_MultiGoal --> [*]

note right of L1_SinglePage
Early Stage:
β=2.0 grounding■■
50% L1 + 40% L2 + 10% L3
end note

note right of L4_MultiGoal
Late Stage:

```

```
α=2.5 progress■■■
10% L1 + 10% L2 + 40% L3 + 40% L4
end note
```

5. ■■■■■■■■■■

5.1 ■■■■■■

[illegible]

5.1.1 Trajectory

SFT ██████████ JSON ██████████

```
"task_id": "demo_001",
"task_goal": "■■■■■ iPhone 15 ■■■■■",
"difficulty_level": 2,
"success": true,
"steps": [
{
"observation": "■■ ■■ ■■ ■■■ ■■■",
"thought": "■1■: ■■■■click■■■■■■",
"action": "click",
"params": {
"element_text": "■■■",
"xpath": "//input[@placeholder='■■']"
}
}
]
}
```

5.1.2 ■■■■■Progress Label■■■

Bootstrap

```
{
  "text": "■■: ■■■■ iPhone 15 ■■■■\n■■: ■■ ■■ ■■",
  "progress": 0.2,
  "step_count": 0,
  "trajectory_id": "demo_001",
  "task_id": "demo_001",
  "outcome": "success"
}
```

5.1.3 ■■■■■■Contrastive Ranking Pair■


```
`margin_ranking_loss` ████████████████████████████████████████████████████
```

****** $\rightarrow$ $\rightarrow$ $\rightarrow$ ******

5.2

5.2.1 LoRA Adapter ■■

[illegible]

- ```
• `adapter_config.json` LoRA r=64 lora_alpha=32 target_modules
• `adapter_model.safetensors`
```

```
step_rl/training/sft_warmup.py:304-305
model.save_pretrained(os.path.join(args.output_dir, "sft_adapter"))
tokenizer.save_pretrained(os.path.join(args.output_dir, "sft_adapter"))
```

```
RL [REDACTED] `PeftModel.from_pretrained()` [REDACTED] Adapter [REDACTED] `merge_and_unload()` [REDACTED] Adapter [REDACTED]
```

### 5.2.2 ■■■■ Checkpoint

[illegible]Checkpoint  `torch.save` [illegible]

### 5.3

### 5.3.1 ■■■■State Memory■

[illegible]

MinHash Simple Hash

- **\*\*Simple Hash\*\*** ■■■ URL + ■ 500 ■■■ MD5 ■■■■■■■■■■
- **\*\*MinHash\*\*** ■■■■ 2-gram shingles ■ 64 ■■■■■■■■■■■■■■■■

```
■■ episode ■■■■ `reset()` ■■■■■■■■■■■■■■■■■■■■■■ novelty ■■■■■■■■■■■■■■■■■■■■■■
```

### 5.3.2 ■■■■Experience Replay■

[illegible]

```
step_rl/training/base_trainer.py:93-95
rb_cfg = config["training"]["replay_buffer"]
self.replay_buffer: Deque[Trajectory] = deque(maxlen=rb_cfg["capacity"])
self.replay_ratio = rb_cfg.get("history_ratio", 0.25)
```

## 5.4

```
flowchart LR
A[] --> B[SFT]
B --> C[]
C --> D[bootstrap_threshold=0.95]
D --> E[RL]
E --> F[]
F --> G[]
G --> E
```

- `data/trajectories/`
- `data/sft/`SFT `.json` / `.jsonl`
- `data/progress/`
- `checkpoints/` Checkpoint 5
- `logs/`
- `outputs/benchmark/`

\*\*

---

## 6.

### 6.1 Web

#### 6.1.1 Playwright

```
`PlaywrightWebEnv`step_rl/environment/playwright_env.pyStep-RLWeb
start()stop()reset()get_observation()execute_action()
`asyncio`
```

```
`async_playwright().start()` Chromium `new_context(viewport=..., java_script_enabled=True)`
episode`reset()`
```

#### 6.1.2 JavaScript DOM

Playwright 1.60+ `page.accessibility`Step-RL JavaScript `page.evaluate()` DOM

```
// step_rl/environment/playwright_env.py:234-261
() => {
const results = [];
const tags = ['a', 'button', 'input', 'textarea', 'select', 'label',
'h1', 'h2', 'h3', 'h4', 'h5', 'h6', 'p', 'span', 'div',
'li', 'td', 'th'];
tags.forEach(tag => {
```

```
document.querySelectorAll(tag).forEach((el, idx) => {
 if (el.offsetParent === null && tag !== 'div') return;
 const rect = el.getBoundingClientRect();
 const text = (el.innerText || el.textContent || el.value || el.placeholder || '').trim();
 const role = el.getAttribute('role') || tag;
 const id = el.id || el.getAttribute('data-testid') || '';
 if (text.length > 0 || id.length > 0 || tag === 'input' || tag === 'button' || tag === 'a') {
 results.push({
 tag: tag, role: role, text: text.slice(0, 200),
 id: id, coords: `${Math.round(rect.x)},${Math.round(rect.y)}`,
 visible: el.offsetParent !== null
 });
 }
});
});
return results;
}
```

17 ID offsetParent === null tag role text id coords  
JS BeautifulSoup HTML

**accessibility snapshot JS 200**

### 6.1.3

```
step_rl/environment/playwright_env.py:138-141
await self._context.route(
 "**/*.{png,jpg,jpeg,gif,svg,css,woff,woff2,ttf}",
 lambda route: route.abort(),
)
```

context.route() abort() 30%–60% 30 5

### 6.1.4

robust\_locate() step\_rl/environment/locator.py:17-94 Multi-attribute Cascade  
Matching element\_id > element\_text tag > xpath > css\_selector > coordinates  
escape\_css\_string() page.locator() document.elementFromPoint()  
data-testid id tag fragile

## 6.2

### 6.2.1 Evidential Learning

ProgressEstimator step\_rl/reward/progress\_estimator.py EvidentialLayer Normal  
Inverse-Gamma NIG

- gamma
- nu precision uncertainty = 1 / nu
- alpha shape > 1
- beta scale

```
step_rl/reward/progress_estimator.py:48-56
h = self.shared(x)
gamma = self.gamma(h)
```

```

nu = F.softplus(self.nu(h)) + 1.0
alpha = F.softplus(self.alpha(h)) + 1.0
beta = F.softplus(self.beta(h)) + 1e-6

```

``softplus``      `+1.0``      ``alpha``      `>`      `1``      ``1``      `/``      `nu``

## 6.2.2

``BaseTrainer._run_episode()``      ``step_rl/training/base_trainer.py:170-178``

```

r_total = (
weights["alpha"] * r_progress * (1.0 - uncertainty)
+ weights["beta"] * r_grounding
+ weights["gamma"] * r_sparse
+ weights["delta"] * r_efficiency
+ weights["epsilon"] * r_novelty
+ weights["zeta"] * r_loop
)

```

`**``      `uncertainty → 1``      `**``      `Uncertainty`  
Gating      reward shaping      `/``

## 6.2.3 Monotonicity Constraint

``monotonicity_loss()``      ``step_rl/reward/progress_estimator.py:278-288``      Hinge Loss

```

diffs = progress_seq[:, 1:] - progress_seq[:, :-1] # [B, T-1]
loss = F.relu(-diffs).mean()
return weight * loss

```

``progress_seq``      ``progress(t+1)``      `<`      `progress(t)``      ``F.relu(-diffs)``  
MSE      Ranking      Evidential NLL

## 6.3 PPO/GRPO

### 6.3.1 Last-Token Log-Prob

Step-RL      LLM      `response``      token

`log-prob``      ``BaseTrainer._policy_forward()``      ``step_rl/training/base_trainer.py:232-291``  
``_get_update_log_probs()``      ``step_rl/training/base_trainer.py:398-446``

```

step_rl/training/base_trainer.py:398-446
def _get_update_log_probs(self, batch_obs, batch_responses):
full_texts = [obs + resp for obs, resp in zip(batch_obs, batch_responses)]
inputs = self.tokenizer(full_texts, return_tensors="pt", padding=True,
truncation=True, max_length=4096)
inputs = {k: v.to(self.device) for k, v in inputs.items()}
outputs = self.policy(**inputs, output_hidden_states=True)
seq_lengths = inputs["attention_mask"].sum(dim=1) - 1
batch_indices = torch.arange(outputs.logits.size(0), device=self.device)
last_logits = outputs.logits[batch_indices, seq_lengths] # [B, vocab]

response_last_tokens = []
for resp in batch_responses:
resp_ids = self.tokenizer(resp, add_special_tokens=False)["input_ids"]
response_last_tokens.append(resp_ids[-1] if resp_ids else self.tokenizer.pad_token_id or 0)
response_last_tokens = torch.tensor(response_last_tokens, device=self.device)

dist = torch.distributions.Categorical(logits=last_logits)
new_log_probs = dist.log_prob(response_last_tokens).float()
return new_log_probs

```

```

prompt response ##### token`seq_lengths = attention_mask.sum() -
1` logits##### Categorical ##### rollout ##### token log-prob##### token
#####`#####`

########## `O(B * L * V)` ##### `B` batch size`L` #####≤4096`V` #####
token##### PPO ##### token log-prob##### 40%

```

## 6.3.2 GAE #####PPO

`PPOTrainer.compute\_gae()`  
#####`step\_rl/training/ppo\_trainer.py:133-161`##### GAE#####Generalized Advantage Estimation#####

```

def compute_gae(self, trajectory):
 rewards = np.array(trajectory.rewards, dtype=np.float64)
 values = np.array(trajectory.values, dtype=np.float64)
 dones = np.array(trajectory.dones, dtype=np.float64)
 T = len(rewards)
 advantages = np.zeros(T, dtype=np.float64)
 returns = np.zeros(T, dtype=np.float64)
 gae = 0.0
 for t in reversed(range(T)):
 if t == T - 1:
 next_value = 0.0
 next_non_terminal = 0.0
 else:
 next_value = values[t + 1]
 next_non_terminal = 1.0 - dones[t]
 delta = rewards[t] + self.gamma * next_value * next_non_terminal - values[t]
 gae = delta + self.gamma * self.gae_lambda * next_non_terminal * gae
 advantages[t] = gae
 returns[t] = gae + values[t]
 advantages = (advantages - advantages.mean()) / (advantages.std() + 1e-8)
 return advantages.tolist(), returns.tolist()

```

#####`γ = 0.99`#####`λ = 0.95`#####GAE#####zero-mean, unit-variance#####

## 6.3.3 GRPO #####

GRPO#####Group Relative Policy Optimization##### Value Model#####group#####

```

step_rl/training/grpo_trainer.py:81-97
def compute_group_advantages(self, trajectories):
 all_advantages = []
 for i in range(0, len(trajectories), self.group_size):
 group = trajectories[i : i + self.group_size]
 returns = [t.total_return for t in group]
 mean_r = np.mean(returns)
 std_r = np.std(returns) + 1e-8
 for t in group:
 A = (t.total_return - mean_r) / std_r
 all_advantages.append([A] * t.length)
 return all_advantages

```

#####`A\_i` = (R\_i - mean) / std`##### Value Model #####`

## 6.3.4 KL #####

PPO ##### GRPO ##### KL #####Kullback-Leibler Divergence#####`PPOTrainer`##### KL #####

```

step_rl/training/ppo_trainer.py:316-321
if self.kl_adaptive:
 if metrics["kl"] > self.kl_target * 2:
 self.kl_coef *= 1.5
 elif metrics["kl"] < self.kl_target / 2:
 self.kl_coef /= 1.5

```

```
self.kl_coef = max(0.01, min(self.kl_coef, 1.0))
```

KL `self.kl_coef` 1.5 1.5 `[0.01, 1.0]` PID "Policy Collapse"

## 6.4

### 6.4.1 MinHash

`StateMemory._minhash()` `step_rl/memory/state_memory.py:81-123`

`n` words `k = 64` permutations

- `O(n * k)` 2-gram shingles `O(n)` shingle `k` `O(n * k)`
- `O(max_states) = O(500)` `OrderedDict`

```
step_rl/memory/state_memory.py:81-123
def _minhash(self, text: str, url: str, num_perm: int = 64) -> str:
 words = text.lower().split()
 if len(words) < 2:
 return self._simple_hash(text, url)
 shingles = set()
 for i in range(len(words) - 1):
 shingles.add(f"{words[i]} {words[i+1]}")
 p = (1 << 61) - 1
 hashes = []
 for seed in range(num_perm):
 a = (seed * 1234567891 + 1) & 0xFFFFFFFFFFFFFFFF
 b = (seed * 9876543211 + 1) & 0xFFFFFFFFFFFFFFFF
 min_val = p
 for s in shingles:
 h = int(hashlib.md5(s.encode()).hexdigest(), 16)
 perm = ((h * a + b) & 0xFFFFFFFFFFFFFFFF) % p
 if perm < min_val:
 min_val = perm
 hashes.append(min_val)
 # ... band folding ...
```

`a`, `b` 64 `band_size = 4`  
LSH `len(words) < 2` `simple_hash` shingle

### 6.4.2

Qwen2.5-7B / Qwen3-8B A100 / L40S GRPO Value Model 30% 8 GB GPU RTX 4060 Laptop RL `config.yaml` `training.algorithm` `grp` `use_4bit: true`

LLM `max_new_tokens=256` A100 1–2 L40S 1.5–2.5 Playwright DOM + 200–500 ms

```
flowchart TB
 subgraph "Web Environment"
 A[Playwright Async
start/stop/reset] --> B[JS DOM Extraction
page.evaluate]
 B --> C[Resource Interception
route.abort]
 C --> D[Robust Locator
multi-attribute cascade]
 end
 subgraph "Reward Shaping"
 E[Progress Estimator
Evidential Layer] --> F[Uncertainty Gating
r * (1 - uncertainty)]
 F --> G[Monotonicity
Hinge Loss]
 end
 subgraph "Policy Optimization"
 H[Rollout
last-token log-prob] --> I{Algorithm}
 I -->|PPO| J[GAE
γ=0.99, λ=0.95]
 I -->|GRPO| K[Group Norm
A=(R-mean)/std]
 end
```

```
J --> L[Adaptive KL
clamp [0.01,1.0]]
K --> L
end
D --> H
G --> H
```

**\*\*Step-RL v2.0** **Web** **Last-Token**  
**Log-Prob** **PPO/GRPO** **\*\***

## 7. DevOps

### 7.1

**Infrastructure** **DevOps** **Development** **and** **Operations** **Step-RL** **v2.0**  
**Containerized** **Deployment** **Docker** **Compose** **Orchestration** **CI/CD**  
**/** **\*\*** **Multi-stage**  
**Build** **\*\***

### 7.2

#### 7.2.1 Dockerfile

**Step-RL v2.0** **Dockerfile** **Multi-stage Build**

```
Stage 1 Builder
`git` `wget` `build-essential` `pip install --user -r requirements.txt` `--user` Python
`playwright` install chromium` Chromium

Stage 2 Runtime
`git` `groupadd -r appgroup && useradd -r -g appgroup appuser`
`root` Builder Python `/root/.local` → /home/appuser/.local` Playwright
`/ms-playwright` → /home/appuser/ms-playwright` `COPY --chown=appuser:appgroup` `root
`EXPOSE 7860 8000` Gradio FastAPI `HEALTHCHECK` 30 `python -c "import
step_rl"
```

**7-1 Dockerfile**

#### 7.2.2

**7-1 Step-RL v2.0**

```
graph TB
 subgraph "Host Machine"
 direction TB
 User[""]
 DataVol[("./data")]
 OutputsVol[("./outputs")]
 ModelsVol[("./models")]
 ConfigVol[("./config.yaml")]
 end

 subgraph "Docker Container"
```





```
graph LR
 subgraph "■■■ CI"
 A[Push/PR] --> B[Python 3.10 ■■]
 A --> C[Python 3.11 ■■]
 B --> D[pytest + coverage]
 C --> D
 D --> E[Codecov ■■]
 D --> F[end_to_end_test.py]
 A --> G[Lint: black/isort/flake8/mypy/bandit]
 end

 subgraph "■■■ CI/CD"
 H[Push to main] --> I[Buildx ■■■■■]
 I --> J[pytest ■■■■■]
 J --> K[Import Check ■■]
 K --> L[Tag v*?]
 L --> M[Docker Hub ■■]
 L --> N[N[■■■■■■]]
 M --> O[■■■■: latest / version / major.minor]
 end

 F --> I
 G --> I

 style D fill:#bfb,stroke:#333,stroke-width:2px
 style M fill:#bfb,stroke:#333,stroke-width:2px
 style G fill:#fbb,stroke:#333,stroke-width:2px
```

### 8.2.1 ■■■■■■URL■■■■■



9.1

9.1.1

Step-RL v2.0 Task Completion Rate 58% 91% Ablation Study Supervised Fine-Tuning, SFT Reinforcement Learning, RL

full\_v2` `grpо` 86~91% Action Grounding Accuracy 95.8% 11.5~13.2 Loop Rate 4~6%

9.1.2

1. `sparse\_ppo` `+progress\_only` 6 Credit Assignment

2. `+grounding\_only` 3% 89.5% 96.5% Action Hallucination

3. GRPO PPO GRPO 86% 91% 4% Group-Relative Advantage Estimation

graph LR

A["SFT Baseline<br/> 58%"] -->|+RL| B["Sparse PP0<br/>68%"]

B -->|+| C["+Progress Only<br/>74%"]

B -->|+| D["+Grounding Only<br/>71%"]

C -->|+| E["Fixed Weight<br/>79%"]

D --> E

E -->|+| F["Full v2 PP0<br/>86%"]

F -->|GRPO| G["GRPO<br/>91%"]

style A fill:#ffcdd2,stroke:#b71c1c

style G fill:#c8e6c9,stroke:#1b5e20

style F fill:#e3f2fd,stroke:#1565c0

9.1.3

`outputs/benchmark\_v2/ablation\_table.md`

full\_v2` Average Return 0.930 Auto-Correction Rate 45% Intervention Rate 13% 5%

9.2

9.2.1 VRAM

Step-RL v2.0 8GB VRAM 7B Video RAM, VRAM

GRPO ██████████ Value Model ██████████ 30% █████ 4-bit █████NF4, Normalized Float 4████████ RTX  
4060 8GB ██████████ GRPO ██████████

## 9.2.2 CPU

CPU GPU CUDA GPU  
RAM 2~4GB

### 9.3 ■■■■■■■■■■

### 9.3.1 ■■■■■

[illegible]

RL 128 rollout + 4 update 15~20 100 epochs 24~36

### 9.3.2 ■■■■■■

■■ 6~12 ■■■■■■■■■■"■■■■■ → ■■■■ → ■■■■ → ■■■"■■■■■■■■■■

```
**Step-RL v2.0 GPU `config.yaml` DeepSpeed
```

— — —

**10.** ■■■■■■

## 10.1 ■■■■■■

■■■■■ Technical Debt ■■■■■ Step-RL v2.0 ■■■■■ Severity ■■■■  
Critical ■■■■■ High ■■■■■ 10 ■■■■■

[illegible]

## 10.2 ■■■■

**XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX**

1. **[Critical]** RL **\*\*** `BaseTrainer` / `PPOTrainer` / `GRPOTrainer` last-token log-prob  
`trl.PPOTrainer` / `trl.GRPOTrainer` KL
2. **[High]** Label Masking **\*\*SFT** prompt masking BPEByte Pair Encoding response  
token <1%
3. **[High]** Replay Buffer Experience Replay PERPrioritized  
Experience Replay — TD Temporal Difference Error — 15~20%

[illegible]

## 10.3 Q1/Q2 Roadmap

```
gantt
 title Step-RL v2.1 / v2.2
 dateFormat YYYY-MM

 section v2.1 (Q1)
 ■■ trl ■■ Trainer :v2_1_1, 2026-01, 2026-03
 ■■ PER ■■■■■■ :v2_1_2, 2026-01, 2026-02
 ■■■■■■ 1B ■■ :v2_1_3, 2026-02, 2026-04
 ■■■■■■■■ + ■■■■ :v2_1_4, 2026-03, 2026-05

 section v2.2 (Q2)
 DeepSpeed / FSDP ■■■■■■ :v2_2_1, 2026-04, 2026-06
 Demo ■■■■■■■■ :v2_2_2, 2026-04, 2026-05
 ■■ Llama / Mistral ■■ :v2_2_3, 2026-05, 2026-07
 ■■ RL ■■■■■■■■ :v2_2_4, 2026-06, 2026-08

 section v2.3 (■■■)
 ■■■■■■ :v2_3_1, after v2_2_4, 2026-10
 ■■■■■■ World Model ■■ :v2_3_2, after v2_3_1, 2026-12
 RLHF for Agents :v2_3_3, after v2_3_2, 2027-03
 ■■■■■■ :v2_3_4, after v2_3_3, 2027-06
```

**■■■■■■■■■■**

- **\*\*v2.1 (Q1)\*\*** [REDACTED] trl [REDACTED] 8GB [REDACTED] 4GB [REDACTED]
- **\*\*v2.2 (Q2)\*\*** [REDACTED] 14B+ [REDACTED] RL [REDACTED]
- **\*\*v2.3 (Q3)\*\*** [REDACTED] RLHF Reinforcement Learning from Human Feedback [REDACTED] for Agents [REDACTED] Web Agent [REDACTED]

[illegible]

## 11

## 11.1 ■ ■ ■

- **\*\*LLM Agent\*\***
- **\*\*RLHF\*\*** Reinforcement Learning from Human Feedback
- **\*\*PPO\*\*** Proximal Policy Optimization
- **\*\*GRPO\*\*** Group Relative Policy Optimization
- **\*\*LoRA\*\*** Low-Rank Adaptation PEFT, Parameter-Efficient Fine-Tuning
- **\*\*GAE\*\*** Generalized Advantage Estimation  $\lambda$
- **\*\*Evidential Learning\*\*** Dirichlet
- **\*\*MinHash\*\***
- **\*\*Grounding\*\***
- **\*\*Curriculum Learning\*\***
- **\*\*SPA\*\*** Single Page Application JavaScript

## 11.2

`config.yaml`

- **\*\*model\*\*** `base\_model` Qwen/Qwen3-8B-Instruct `fallback\_models` `dtype` bf16/fp16/fp32 `use\_4bit` 4-bit
- **\*\*lora\*\*** `r` 64 `lora\_alpha` 32 `target\_modules` q/k/v/o\_proj MLP `dropout` 0.05
- **\*\*environment\*\*** `browser` chromium `headless` `viewport` 1280x720 `blocked\_domains`
- **\*\*curriculum\*\*** `total\_epochs` 100 `levels` / / `promotion\_threshold` 0.90
- **\*\*reward\*\*** `sparse` / / `progress\_estimator` evidential grounding `state\_memory` MinHash `dynamic\_weights` early/mid/late
- **\*\*training\*\*** `algorithm` grpo/ppo `ppo` PPO `clip\_range` kl\_coef gae\_lambda vf\_coef `grpo` GRPO `group\_size` clip\_range kl\_coef `sft` SFT Warmup 2e-4 epoch 3 4 `replay\_buffer` 10000 alpha/beta
- **\*\*evaluation\*\*** `num\_episodes` 100 `metrics` 10 `ablation\_studies` 9
- **\*\*checkpoint\*\*** `save\_dir` `keep\_last\_n` 5 `auto\_resume`
- **\*\*demo\*\*** `gradio\_port` 7860 `api\_port` 8000 `enable\_human\_feedback`
- **\*\*continual\*\*** `enabled` `bootstrap\_threshold` 0.95 `retrain\_interval\_episodes` 1000

## 11.3

`requirements.txt`

- **\*\*Python 3.10+\*\*** 3.10
- **\*\*torch>=2.1.0\*\*** `transformers>=4.40.0` `accelerate>=0.28.0` `datasets>=2.18.0`
- **\*\*RL\*\*** `peft>=0.10.0` LoRA `trl>=0.8.0` RL `bitsandbytes>=0.43.0` 4-bit/8-bit
- **\*\*Web\*\*** `playwright>=1.43.0` `beautifulsoup4>=4.12.0` HTML `lxml>=5.1.0` XML
- **\*\*PyYAML>=6.0\*\*** `numpy>=1.26.0` `pandas>=2.2.0` `scikit-learn>=1.4.0` `pillow>=10.2.0` `imagehash>=4.3.1` `tqdm>=4.66.0` `wandb>=0.16.0` `matplotlib>=3.8.0` `seaborn>=0.13.0` `tabulate>=0.9.0`
- **\*\*Demo API\*\*** `gradio>=4.25.0` `fastapi>=0.110.0` `uvicorn>=0.29.0`
- **\*\*pytest>=8.1.0\*\*** `pytest-asyncio>=0.23.0` `black>=24.0.0` `isort>=5.13.0` `flake8>=7.0.0`
- **\*\*deepspeed>=0.14.0\*\*** GPU

## 11.4 Checklist

1. **\*\*Python 3.10+\*\*** CUDA 11.8+ GPU 8GB+ VRAM GRPO + 4-bit
2. **\*\*pip install -r requirements.txt && playwright install chromium\*\*** Python
3. **\*\*python scripts/prepare\_mock\_data.py\*\*** SFT 126 41

```

4. **██████████** `pytest tests/ -v ████████ **52 ██████████** `python scripts/end_to_end_test.py` ████████████████

5. **SFT ██████** `python -m step_rl.training.sft_warmup --config config.yaml --data_dir ./data/sft --output_dir ./outputs/sft_ecommerce --use_4bit` █

6. **██████████████** `python -m step_rl.reward.train_reward_model --config config.yaml --data_path ./data/progress/ecommerce_labels.json --output_dir ./checkpoints/progress_estimator --use_uncertainty yes` █

7. **GRPO/PPO ██████** `python -m step_rl.training.grpo_trainer --config config.yaml --sft_adapter ./outputs/sft_ecommerce/sft_adapter --progress_model ./checkpoints/progress_estimator/best_model.pt --output_dir ./checkpoints/grpo` █

8. **██████** `python -m step_rl.evaluation.benchmark --config config.yaml --mock`██ 10 ████████████████

9. **Demo ██████** `python -m step_rl.demo.demo --config config.yaml --policy ./checkpoints/grpo/best_adapter`████
`http://localhost:7860` ██ Gradio ██████

10. **Docker ██████** `docker build -t step-rl:latest` ████ `docker-compose --profile demo up` ████████ `docker-compose --profile train up -d` ██████

```

[illegible]